

Understanding Customer Sentiment on Twitter

An NLP analysis of tweets about Apple & Google products

The Business Problem

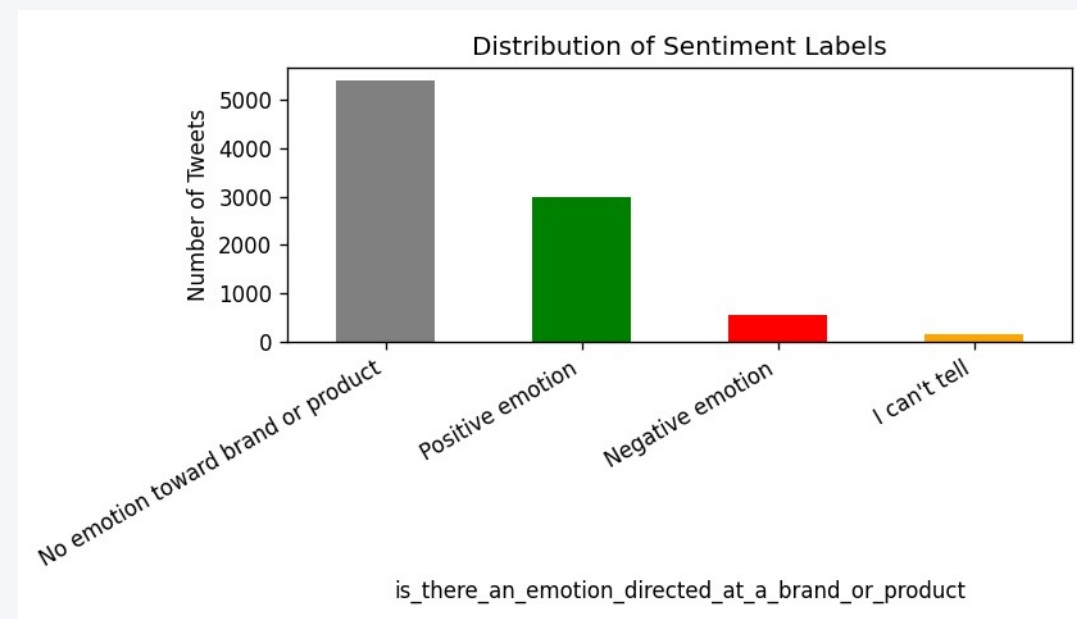
Brands need to know how people feel about their products — fast

- Thousands of tweets are posted about Apple and Google products every day
- Reading every tweet by hand to gauge public reaction is not realistic
- Marketing and product teams need a fast, automated way to know: is the public reaction positive, negative, or neutral?
- Goal: build a model that reads a tweet and classifies the sentiment automatically

The Data

~9,000 real tweets, hand-labeled for sentiment

- Source: tweets collected during a major tech event (SXSW), mentioning Apple and Google products
- Each tweet was hand-labeled by a human as Positive, Negative, or No emotion
- Most tweets (about 6 in 10) show no clear emotion — they're just factual or informational
- About 1 in 3 tweets are positive, while only about 1 in 16 are clearly negative
- This imbalance matters: negative tweets are rare, but they're the ones a brand most needs to catch



Preparing the Tweets

Cleaning the text so the model can focus on what matters

- Converted all text to lowercase
- Removed links, @usernames, and # symbols — these don't carry sentiment
- Removed punctuation and numbers
- Removed common filler words ("the", "and", "is"...) so the model focuses on meaningful words
- Result: clean, simple text ready to be turned into numbers the model can learn from

Comparing Three Models

From a simple baseline to more advanced approaches

Naive Bayes

A simple, fast baseline model used as a starting point for comparison.

Logistic Regression

A widely-used model that weighs words to predict sentiment — and tells us how confident it is.

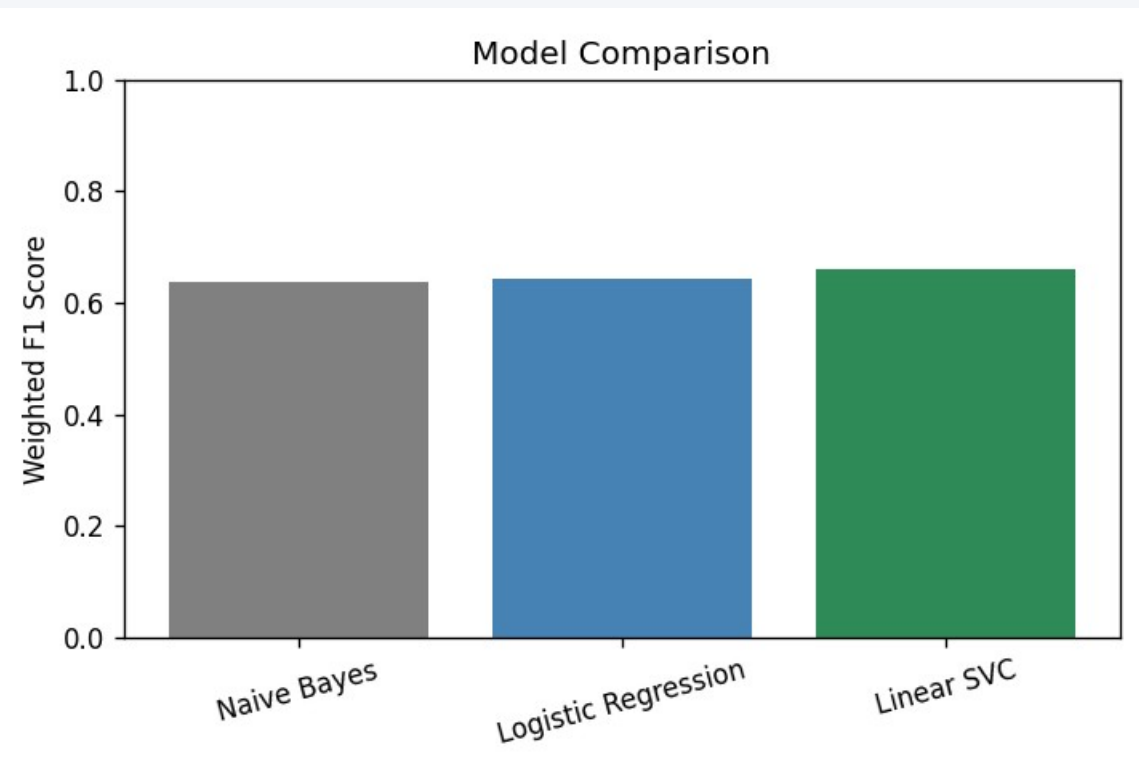
Linear SVC

A more advanced model that draws clear boundaries between sentiment categories.

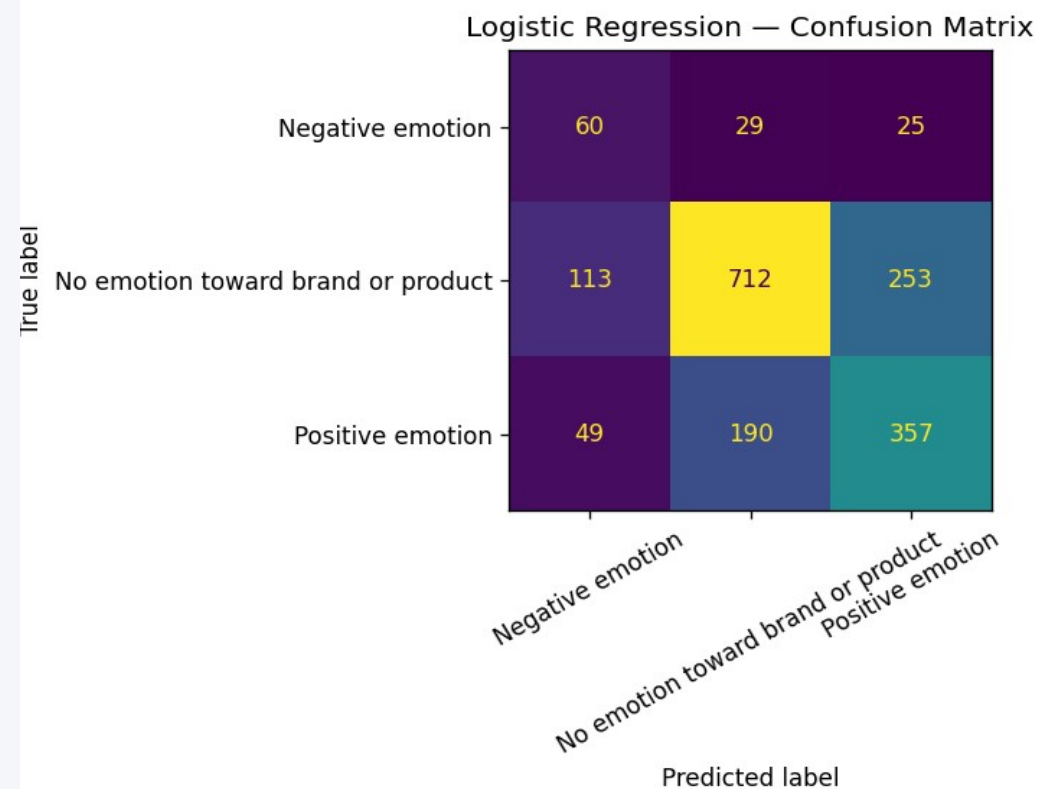
All three models were trained on 80% of the tweets and tested on the remaining 20% they had never seen before.

Results

Logistic Regression and Linear SVC both outperform the baseline



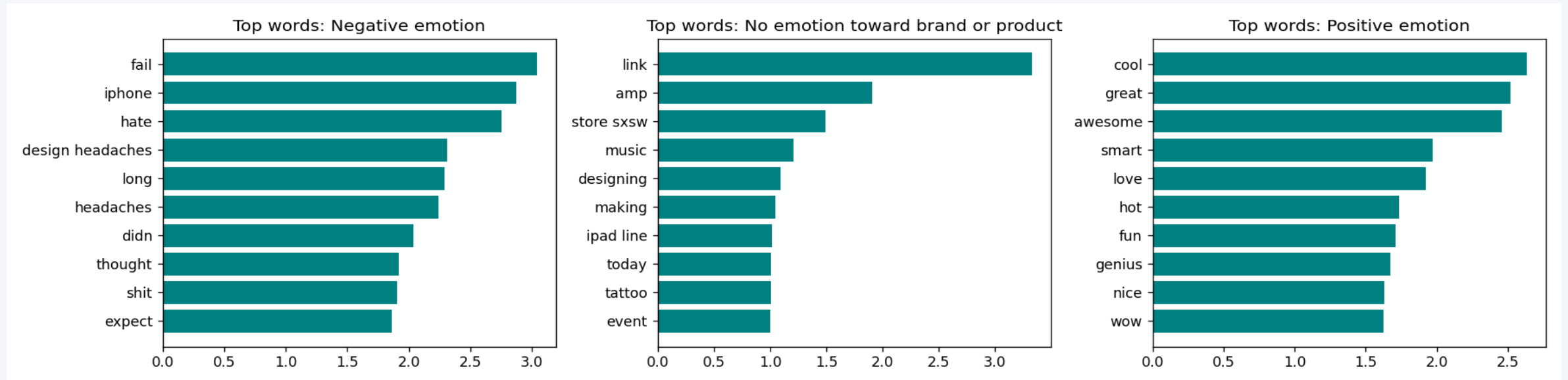
Logistic Regression was chosen as the final model — it performs as well as the more complex SVC, while also telling us how confident it is in each prediction.



Can We Trust the Model?

Looking inside the model's decisions

- Using a technique called LIME, we can see exactly which words drove each prediction
- Positive predictions are driven by words like "great", "love", and "awesome"
- Negative predictions are driven by words like "hate", "fail", and "broken"
- This matches human intuition — the model has learned genuine sentiment signals, not noise



Recommendations & Next Steps

- **Use it as a first pass, not a final answer**

Let the model sort incoming tweets, then have a person review anything flagged as negative.

- **Prioritize catching negative tweets**

Missing a complaint is costlier than double-checking a false alarm — tune the model to favor catching negatives.

- **Collect more negative examples over time**

Negative tweets are rare in this dataset; more examples will help the model get even better at spotting them.

- **Consider more advanced language models later**

Newer AI language models could better handle sarcasm and subtle tone in future versions.